

Commutativity Violations and Contingent Attitudes: How Early Interaction Shapes Future Beliefs

Appendix

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This Appendix provides a comprehensive overview of the computer simulations supporting the analyses in the paper. Section A gives an overview of the parameter space that we explored via computer simulations and directs the reader to the relevant summary plots contained in the Supplementary Materials.¹ Section B discusses the notion of an *attitude* about H as introduced in Section 4 of the article, and shows that our arguments in the paper remain valid if one uses alternative notions of attitude.

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¹All supplementary materials can be found at the link <https://github.com/LillyPHIL/Commutativity-Violations-and-Contingent-Attitudes-How-Early-Interaction-Shapes-Future-Beliefs.git>

Variable	Assignments
H	$\Pr(H) \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$
N_S	permutations over $\{tt, ft, rand\}$ of the triples (0.1, 0.1, 0.8), (0.2, 0.3, 0.5), (0.1, 0.2, 0.7), (0.3, 0.3, 0.4)
$B_S H \cdot N_S$	Five Scenarios in Table 2
q	0.7
Report sequence length	300

Table 1: Overview of the parameter space explored via computer simulations ($5 \times 18 \times 5 \times 1 \times 1$).

A Experiment Overview

In this section, we overview the computer simulations we conducted. The code for the simulation as well as the code for data analysis is available in the Supplementary Material folder. Generative AI tools (OpenAI Chat GPT v.5.6, Anthropic Claude Sonnet 5) were used to assist with code development, debugging, and revision. Additionally, they were used to assist data analysis and plotting (Anthropic Claude Sonnet 5).

An overview of the explored parameter space of the model described in Section 5 of the paper is given in Table 1. The variable H is initialised by fixing a value for the probability that $H = H$; the variable N_S is initialised by fixing a particular probability distribution over its values $\{tt, ft, rand\}$; the conditional probability table for B_S is determined by one of the five scenarios in Table 2; the probability that a source provides a positive report is the probability $q = 0.7$ that a positive report is drawn from a Bernoulli distribution; report sequences of length 300 are considered. The technical details of how each considered Scenario is initialised are provided in Table 2.

For every generated report sequence, we recorded the value of the following variables:

V_{BS}	tt		ft		r	
	H	$\neg$ H	H	$\neg$ H	H	$\neg$ H
.1	0	.25	.25	0	0	0
.2	0	.25	.25	0	0	0
.3	0	.25	.25	0	0	0
.4	0	.25	.25	0	0	0
.5	0	0	0	0	1	1
.6	.25	0	0	.25	0	0
.7	.25	0	0	.25	0	0
.8	.25	0	0	.25	0	0
.9	.25	0	0	.25	0	0

(a) Scenario 1: No Expectations

V_{BS}	tt		ft		r	
	H	$\neg$ H	H	$\neg$ H	H	$\neg$ H
.1	0	.1	.1	0	0	0
.2	0	.2	.2	0	0	0
.3	0	.3	.3	0	0	0
.4	0	.4	.4	0	0	0
.5	0	0	0	0	1	1
.6	.4	0	0	.4	0	0
.7	.3	0	0	.3	0	0
.8	.2	0	0	.2	0	0
.9	.1	0	0	.1	0	0

(c) Scenario 3: Erratic Reporting

V_{BS}	tt		ft		r	
	H	$\neg$ H	H	$\neg$ H	H	$\neg$ H
.1	0	.4	.4	0	0	0
.2	0	.3	.3	0	0	0
.3	0	.2	.2	0	0	0
.4	0	.1	.1	0	0	0
.5	0	0	0	0	1	1
.6	.1	0	0	.1	0	0
.7	.2	0	0	.2	0	0
.8	.3	0	0	.3	0	0
.9	.4	0	0	.4	0	0

(b) Scenario 2: Consistent Reporting

V_{BS}	tt		ft		r	
	H	$\neg$ H	H	$\neg$ H	H	$\neg$ H
.1	0	.4	.1	0	0	0
.2	0	.3	.2	0	0	0
.3	0	.2	.3	0	0	0
.4	0	.1	.4	0	0	0
.5	0	0	0	0	1	1
.6	.1	0	0	.4	0	0
.7	.2	0	0	.3	0	0
.8	.3	0	0	.2	0	0
.9	.4	0	0	.1	0	0

(d) Scenario 4: As Good As They Come

V_{BS}	tt		ft		r	
	H	$\neg$ H	H	$\neg$ H	H	$\neg$ H
.1	0	.1	.4	0	0	0
.2	0	.2	.3	0	0	0
.3	0	.3	.2	0	0	0
.4	0	.4	.1	0	0	0
.5	0	0	0	0	1	1
.6	.4	0	0	.1	0	0
.7	.3	0	0	.2	0	0
.8	.2	0	0	.3	0	0
.9	.1	0	0	.4	0	0

(e) Scenario 5: Beware Consistency

Table 2: Technical Details of the Five Scenarios for B_S Considered in the Paper.

- Generated sequence;
- For each processing type (global, sequential static, sequential dynamic):
 - Posterior probability of H given the whole sequence;
 - Posterior probability of tt given the whole sequence;
 - Posterior probability of ft given the whole sequence;

- Posterior probability of *rand* given the whole sequence;
- Posterior probability of each behaviour b_j of B_S given the whole sequence.

The Supplementary Materials folder contains four summary plots. We list them here below with a brief description of their content:

- *supp_01_all_anchoring_dynamic.jpg* plots for the full parameter space the probability of forming an attitude favouring H for dynamic sequential processing against prior for H and two different assignments for N_S , and each scenario (each point is a proportion computed over 1000 report sequences and is associated with a corresponding 95% confidence interval);
- *supp_02_all_anchoring_static.jpg* plots for the full parameter space the probability of forming an attitude favouring H for static sequential processing against prior for H and two different assignments for N_S , and each scenario (each point is a proportion computed over 1000 report sequences and is associated with a corresponding 95% confidence interval);
- *supp_03_all_contingency_dynamic.jpg* plots for the full parameter space the probability of different attitudes towards H for dynamic sequential processing against prior for H , the different assignments for N_S and the distinct scenarios (each data point is a proportion computed over 1000 report sequences and is associated with a 95% confidence interval);
- *supp_04_all_contingency_static.jpg* plots for the full parameter space the probability of different attitudes towards H for static sequential processing against prior for H , the different assignments for N_S and the distinct scenarios (each data point is a proportion computed over 1000 report sequences and is associated with a 95% confidence interval);

We conclude this section with a formal remark. The reader will notice that for the first three scenarios, whenever N_S is initialised such that $\Pr(tt) = \Pr(ft)$, and $\Pr(H) = 0.5$, both static and dynamic sequential processing necessarily generate an attitude favouring neither H nor $\neg H$, which corresponds to one's posterior in H being equal to 0.5. We explain this as follows. Recall the notation introduced in Section 2 of the article. In the terminology of Baccini and Hartmann [2026], Scenarios 1, 2, 3 are *source-symmetric*, i.e., for all b_j in the range of B_S , it is simultaneously the case that (i) $\Pr(b_j|H \cdot tt) = \Pr(b_j|\neg H \cdot ft)$ and (ii) $\Pr(b_j|\neg H \cdot tt) = \Pr(b_j|H \cdot ft)$. It is also shown in Baccini and Hartmann [2026, Proposition 3 (ii)] that if $\Pr(tt) = \Pr(ft)$, then one's globally computed posterior belief in H $\Pr^{(k,n)}(H)$ is equal to the prior $\Pr(H)$ for any k and n . Additionally, it is easy to show that in the same situation, one's globally computed posteriors $\Pr^{(k,n)}(tt)$ and $\Pr^{(k,n)}(ft)$ will also be equal to each other. Using the properties of Bayesian networks (see, e.g., Neapolitan [2003], Hartmann [2021]), one can easily compute the posterior probabilities for tt and ft in the Baccini and Hartmann [2026] model upon learning any set of $n > 0$ reports of which exactly k are positive:

$$\Pr^{(k,n)}(tt) = \frac{t}{t + \frac{f(h x_{k,n}^{H,ft} + \bar{h} x_{k,n}^{\neg H,ft}) + r x_{k,n}^{H,rand}}{h x_{k,n}^{H,tt} + \bar{h} x_{k,n}^{\neg H,tt}}}$$

$$\Pr^{(k,n)}(ft) = \frac{f}{f + \frac{t(h x_{k,n}^{H,tt} + \bar{h} x_{k,n}^{\neg H,tt}) + r x_{k,n}^{H,rand}}{h x_{k,n}^{H,ft} + \bar{h} x_{k,n}^{\neg H,ft}}}$$

where recall that for $hyp \in \{H, \neg H\}$, $rel \in \{tt, ft, rand\}$,

$$x_{k,n}^{hyp,rel} := \sum_{b_j \in V_{B_S}} a_{b_j}^{hyp,rel} \cdot b_j^k \cdot (1 - b_j)^{n-k}.$$

It follows straightforwardly from the definition of *source-symmetry* that for any k and any

n , $x_{k,n}^{H,tt} = x_{k,n}^{\neg H,ft}$ and $x_{k,n}^{\neg H,tt} = x_{k,n}^{H,ft}$. With the equalities above one obtains that

$$\Pr^{(k,n)}(tt) = \frac{t}{t + \frac{f(h x_{k,n}^{\neg H,tt} + \bar{h} x_{k,n}^{H,tt}) + r x_{k,n}^{H,\text{rand}}}{h x_{k,n}^{\neg H,ft} + \bar{h} x_{k,n}^{H,ft}}}.$$

By the fact that $h = \bar{h}$ and that $t = f$,

$$\Pr^{(k,n)}(tt) = \frac{f}{f + \frac{t(\bar{h} x_{k,n}^{\neg H,tt} + h x_{k,n}^{H,tt}) + r x_{k,n}^{H,\text{rand}}}{\bar{h} x_{k,n}^{\neg H,ft} + h x_{k,n}^{H,ft}}} = \Pr^{(k,n)}(ft).$$

This is sufficient to prove that if a set of reports is processed globally, whenever $\Pr(H) = 0.5$ and $\Pr(tt) = \Pr(ft)$, then $\Pr^{(k,n)}(H) = \Pr(H) = 0.5$ and $\Pr^{(k,n)}(tt) = \Pr^{(k,n)}(ft)$. Since this holds for any k, n , it also holds when $n = 1$.

Notice that sequential processing consists of a sequence of m updates whereby at each time-step only one report is processed, and therefore, at each time-step, $n = 1$. We can then show that also for sequential processing, both dynamic and static, one's posterior in H remains 0.5 throughout the update process, provided that initially $\Pr(H) = 0.5$ and $\Pr(tt) = \Pr(ft)$. To show this it is sufficient to prove that source-symmetry is preserved stepwise after update on a single report.

We show this point distinctly for static sequential processing and dynamic sequential processing. Note that static sequential processing trivially preserves source-symmetry, since the B_S table is not updated: Hence, whenever the conditional table for B_S satisfies source symmetry, then it will satisfy it after every update. For dynamic update instead, one needs to check that for every b_j in the range of B_S it is the case that (a) $\Pr(b_j|H \cdot tt \cdot \text{Rep}) = \Pr(b_j|\neg H \cdot ft \cdot \text{Rep})$ and (b) $\Pr(b_j|\neg H \cdot tt \cdot \text{Rep}) = \Pr(b_j|H \cdot ft \cdot \text{Rep})$. We illustrate how to prove that (a). Analogous arguments can be given for (b), and they are sufficient to prove inductively that any sequence processed sequentially preserves source symmetry under the initial assumptions that $\Pr(H) = 0.5$ and $\Pr(tt) = \Pr(ft)$.

By the rules of probability and the properties of Bayesian networks, one obtains that:

$$\Pr(b_j | H \cdot tt \cdot Rep) = \frac{\Pr(b_j \cdot H \cdot tt \cdot Rep)}{\Pr(H \cdot tt \cdot Rep)} = \frac{\Pr(H) \cdot \Pr(tt) \cdot \Pr(b_j | H \cdot tt) \cdot \Pr(Rep | b_j)}{\sum_{b_j} \Pr(H) \cdot \Pr(tt) \cdot \Pr(b_j | H \cdot tt) \cdot \Pr(Rep | b_j)}$$

By the fact that $\Pr(tt) = \Pr(ft)$ and that $\Pr(H) = 0.5$, and that the B_S table satisfies source symmetry, we can rewrite the expression above as

$$\frac{\Pr(\neg H) \cdot \Pr(ft) \cdot \Pr(b_j | \neg H \cdot ft) \cdot \Pr(Rep | b_j)}{\sum_{b_j} \Pr(\neg H) \cdot \Pr(ft) \cdot \Pr(b_j | \neg H \cdot ft) \cdot \Pr(Rep | b_j)}$$

which equals $\Pr(b_j | \neg H \cdot ft \cdot Rep)$. This is sufficient to show (a).

In light of these mathematical findings, we corrected for floating point errors causing deviations from a posterior in H not equal to 5 in the first three scenarios whenever $\Pr(H) = 0.5$ and $\Pr(tt) = \Pr(ft)$.

B The Concept of Attitude

In this section, we discuss the concept of attitude introduced in Section 4 of the paper. We begin by observing that, upon iterated interaction, sequential processing generally leads one to entertain posterior degree of belief in H either very close to 0 or to 1. Figure 1 plots how distant one's posterior degree of belief in H is from 0.5, after processing 300 reports from a source who reports positively with probability 0.7. Each data point corresponds to a posterior computed over a particular report sequence (1000 sequences per parameter combination). The plots reports all posteriors for H for all N_S prior assignments sorted by scenarios. Figure 1 makes clear that both kind of sequential processing tend to lead one to entertain beliefs close to 1 or 0 (most points are located around a distance of 0.5 from 0.5). We notice however that static sequential update differs from dynamic sequential update in that it admits in several cases that one's posterior for H be close to 0.5. Thus, one may wonder whether the way

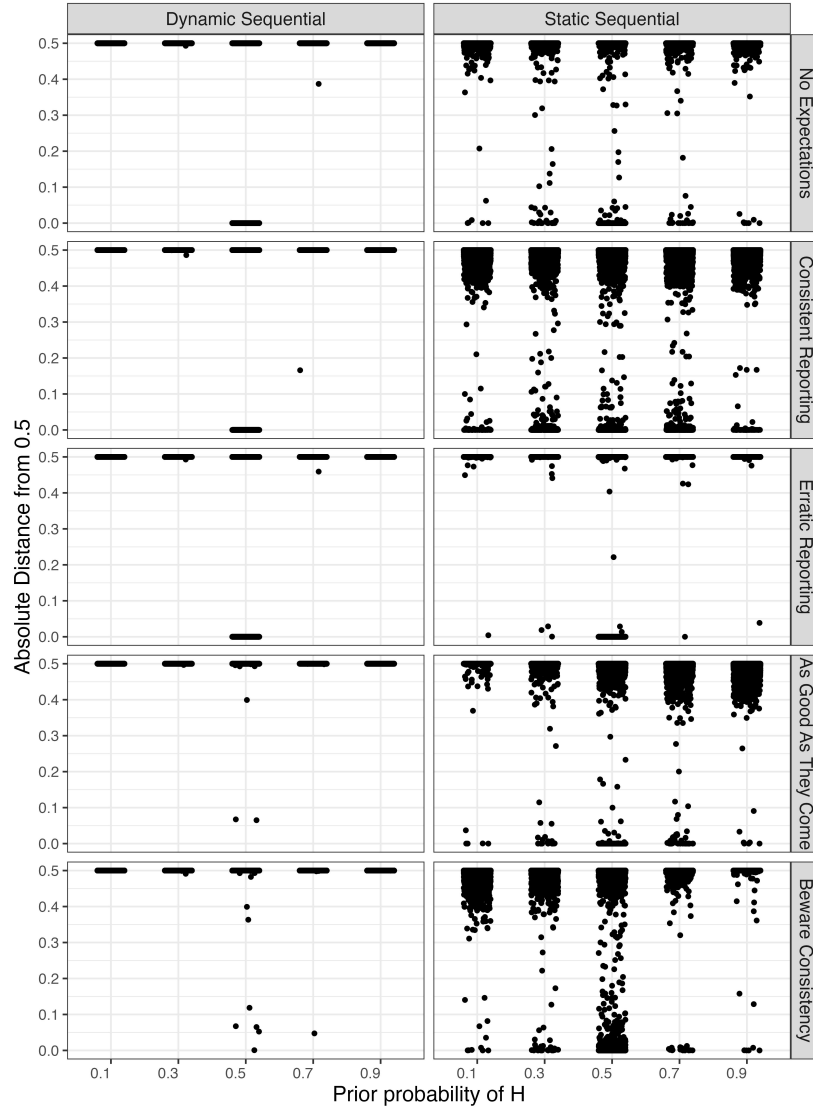


Figure 1: Distribution of Posteriors for H for static (right) and dynamic (left) sequential processing, against prior for H (x-axis) across different scenarios (top facet titles). Each point is a value output on the basis of 300 reports generated by a source reporting positively with probability 0.7.

we defined attitudes towards H is justified, and whether the arguments put forth in Sections 6 and 7 crucially depend on the particular way in which we defined an agent's attitude towards H . We show that this is not the case.

Effectively, an attitude towards an issue can be determined with the following construction. First, one partitions the $[0, 1]$ posterior degree of beliefs interval into three distinct

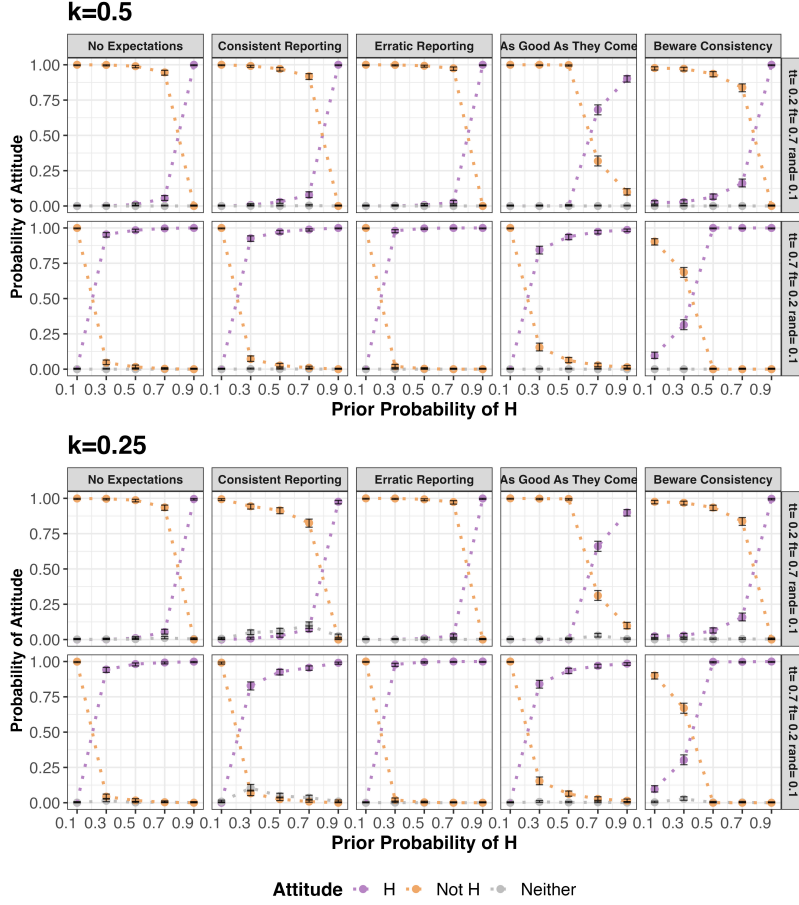


Figure 2: Probability of different attitudes towards H (bottom legend, y-axis) for static sequential processing for $k = 0.5$ (top) and $k = 0.25$ (bottom), against prior for H (x-axis) and two different assignments for N_S (right facet titles) across different scenarios (top facet titles). Each point is a proportion computed over 1000 sequences of 300 reports generated by a source reporting positively with probability 0.7. Each point is associated with a corresponding 95% confidence interval.

intervals $\{[0, k[, [k, 1 - k],]1 - k, 1]\}$ where $k \in]0, 1[$. Then, one defines a function $f : [0, 1] \rightarrow \{H, \neg H, \emptyset\}$ such that for all $i \in [1, 0]$, $f(i) = H$ if $i > 1 - k$, $f(i) = \neg H$ if $i < k$, and $f(i) = \emptyset$ otherwise. From this viewpoint, the definition of attitude given in Section 4 in the article assumes that $k = 0.5$: If one's posterior degree of belief for H is bigger than 0.5, then one favours H ; if it is smaller than 0.5 then one favours $\neg H$; if it is exactly 0.5, then the agent favours neither. What happens if one sets $k < 0.5$, thus modelling agents who more

often refrain from favouring any value for H ? Already from Figure 1, in the case of dynamic processing, since output posteriors for H are generally close to either 0 or 1, setting k to a smaller value does not widely affect how posteriors for H are mapped to values for H , unless k is extremely small (arguably corresponding to an unrealistic case where agents do not form attitudes unless they are almost fully certain of it). As opposed to that, we see that smaller but reasonable values for k may interestingly affect how posterior degrees of belief for H are mapped to attitudes in the case of static sequential processing. In such cases, how would a different k affect the analyses in Section 6 and 7 of the article? Figures 2 and 3 show that the analyses in Section 6 and 7 of the article remain valid, even if k is set lower than 0.5.

First, in light of Figure 1, it seems that the output posteriors for H for static sequential processing may be divided into two groups depending on whether they are closer to 0.5 or the extremes 0, 1. Such a classification seem to be adequately implementable by fixing $k = 0.25$. This corresponds to coming to favour H ($\neg H$) whenever one’s posterior is above 0.75 (below 0.25), and refraining from favouring either H or $\neg H$ otherwise.

Figure 2 makes clear that across all scenarios, different values for k do not affect the diagnosis we put forth in Section 6 of the article: Exactly in those instances where an agent is surprised by the source’s report, if and what conclusion is drawn remains contingent, as signalled by the fact that in those cases none of the final state of an agent is uniquely determined. Instead, what could change by reducing k is the likelihood of coming to hold a particular attitude.

Finally, Figure 3 shows that lowering k does not affect the arguments given in Section 7 of the article: The first item in a sequence of observations determines different likelihoods of forming a particular attitude; the extent to which one expects a source to be consistent modulates the weight one assigns to the first acquired report.

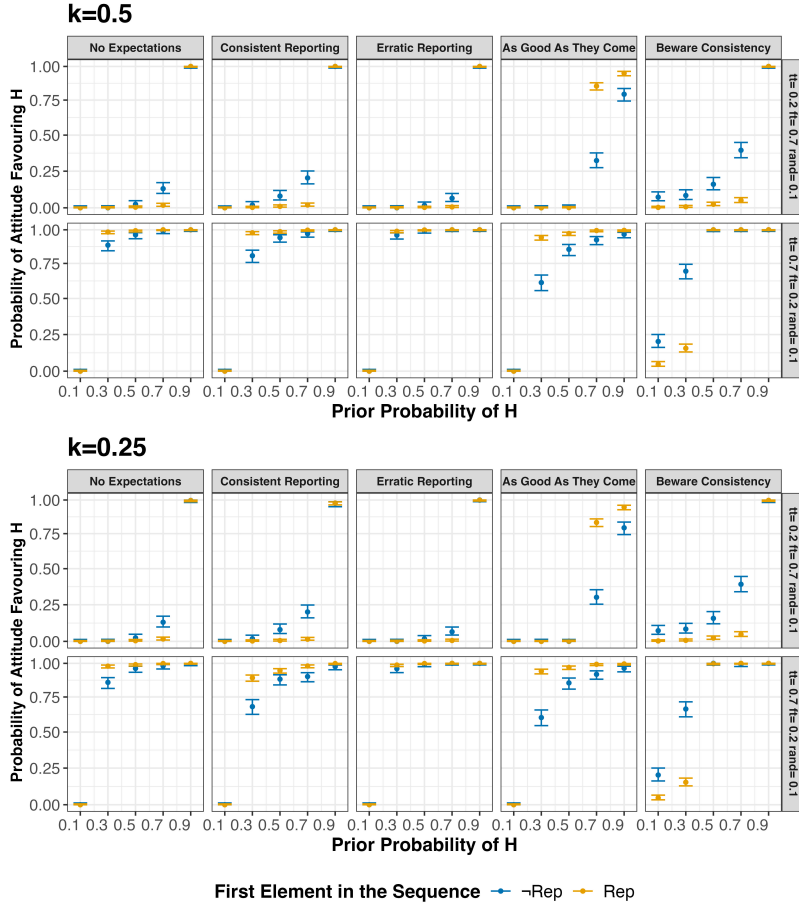


Figure 3: Probability of forming an attitude favouring H (y-axis) for static sequential processing for $k = 0.5$ (top) and $k = 0.25$ (bottom), against prior for H (x-axis) and two different assignments for N_S (top facet titles) across scenarios (top facet titles). Each point is a proportion computed over 1000 sequences of 300 reports generated by a source reporting positively with probability 0.7. Each point is associated with a corresponding 95% confidence interval.

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